Probability

Topic 6: Basic probability

QUESTION 1 A card is drawn at random from a normal pack of 52 cards. Find the probability that the card is

a a diamond ______________ **b** a red card ______________ **c** a king _____________

d not a club ______________ **e** a red ace ______________ **f** a black card _____________

QUESTION 2 From the letters of the word FREQUENCY, one letter is selected at random. What is the probability that the letter is

a a vowel? ______________________________ **b** a consonant? ___________________________

c the letter R? ___________________________

QUESTION 3 A die is thrown once. Find the probability that the number is

a a six _______________________________ **b** an even number _________________________

c a number less than 5 ___________________ **d** seven _________________________________

e a prime number ___________________ **f** a square number ________________________

QUESTION 4 A bag contains 8 white, 5 yellow and 7 blue balls. If a ball is drawn at random, find the probability that it is

a white ______________ **b** yellow ______________ **c** blue _____________

d not white ______________ **e** red ______________ **f** either white or blue _______

QUESTION 5 A three-digit number is to be formed from the digits 4, 5 and 6 written on separate cards. What is the probability that the number formed is

a odd? _________________________ **b** even? _________________________

c less than 600? _________________________ **d** divisible by 3? _______________________

e divisible by 5? ________________________ **f** greater than 600? ______________________

QUESTION 6 The numbers 1 to 9 are written on separate cards. One card is chosen at random. What is the probability that

a the number is even? _______________________ **b** the number is odd? _____________________

c it is 5? _________________________ **d** it is 10? _______________________

e it is a prime number? _____________________ **f** it is divisible by 3? ______________________

QUESTION 7 A letter is chosen from the word EXCELLENT. What is the probability that the letter is

a a vowel? _______________________ **b** a consonant? _____________________

c the letter L? _________________________ **d** the letter N? _______________________

e the letters E or L? _____________________ **f** the letters T or X? ______________________

Probability

Topic 7: Simple probability

QUESTION **1** Complete the following sentence.

The probability of any event is always in the range from ____________ to ____________.

QUESTION **2** A bag contains 6 red marbles and 2 green marbles. If one marble is drawn at random, what is the probability, as a decimal, that it is

a red? ______________ **b** green? ______________ **c** white? ______________

QUESTION **3** A raffle ticket is drawn from a box containing 100 tickets numbered from 1 to 100. Find the percentage chance that the number of the ticket is

a divisible by 10 ______________________

b less than 10 ______________________

c greater than 10 ______________________

d a multiple of 5 ______________________

e greater than 90 ______________________

f a number containing the digit 9 ____________

QUESTION **4** A spinner used in a game is in the shape of a pentagon and has an equal chance of landing on any of its sides. The sides are numbered 1, 2, 3, 4 and 5. What is the probability, as a percentage, that the spinner lands on

a 5? ______________________________

b an even number? ______________________

c an odd number?______________________

d a number greater than 3? ________________

QUESTION **5** A bag holds 9 brown, 7 yellow and 4 white golf balls. If a ball is selected at random from the bag, what is the probability (as a fraction in its simplest form) that the ball is

a brown? __________________________

b yellow? ____________________________

c white? __________________________

d brown or yellow? ______________________

e pink? __________________________

f brown, yellow or white? _________________

g yellow or white? ______________________

h white or brown? ______________________

i not white? ______________________

j neither yellow nor brown nor white? ________

